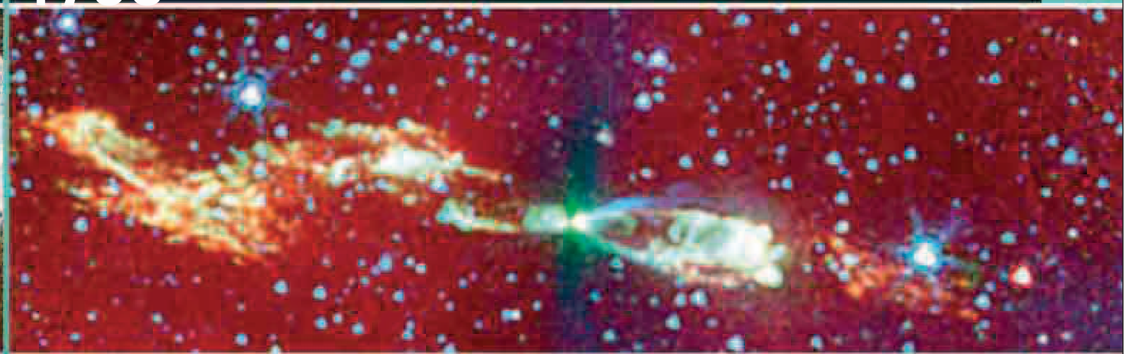
1796

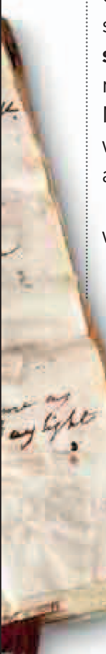


An image from the Spitzer space telescope reveals a distant star and planets forming from a spinning gas cloud, in the same way that our Solar System was created billions of years ago.

eyes that was robbing him of the ability to see green or red. He requested that his eyes be examined after his death to prove this thesis, but the fluid was normal and clear. A DNA test on his preserved eyes in the 1990s showed that he had a common genetic deficiency that reduced his sensitivity to green, proving **he did indeed have color blindness**.

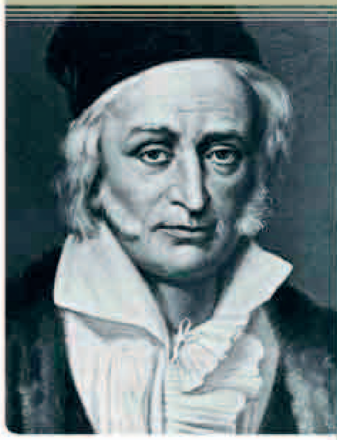
Up to this point in time, meteorites were thought to come from volcanoes, but in 1794, German physicist **Ernst Chladni** (1756–1827) made the suggestion that **they came from space**. The following year a large meteorite fell to earth at Wold Newton in Yorkshire, England, which was far from any volcano and backed up Chladni's theory.

In 1795, the French government was championing a **new decimal system of measures**. They decreed that a gram was "the absolute weight of a volume of water equal to the cube of the hundredth part of the meter, at the temperature of melting ice."



Dalton's color blindness test
Dalton used this book of colored threads to test his own color blindness. He suffered from inherited red-green color blindness.

CARL FRIEDRICH GAUSS (1777–1855)



Born in the German Duchy of Brunswick, Gauss was a mathematical genius who astonished people with his brilliance at an early age. He grew up to be one of the greatest mathematicians ever, making important contributions to the theory of numbers, non-Euclidean geometry, planetary astronomy, probability, and the theory of functions.

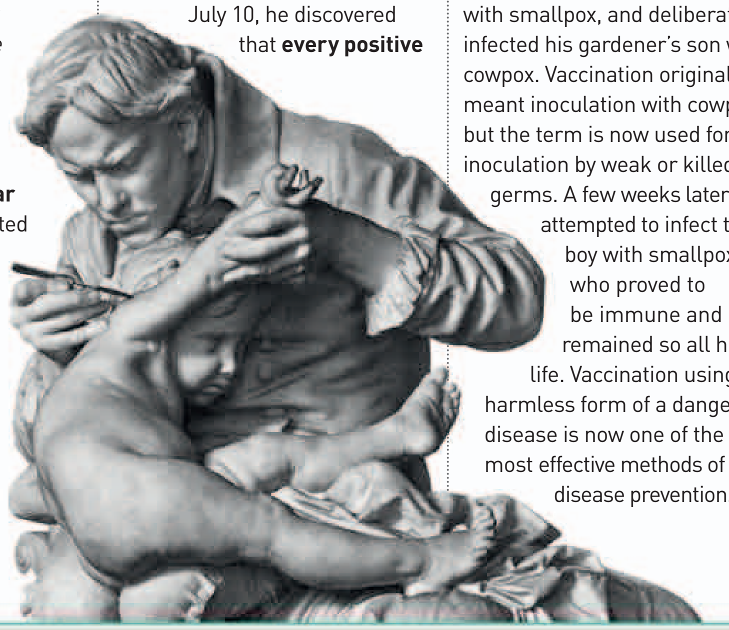
Also in France, the naturalist **George Cuvier** (1769–1832) introduced the idea that **Earth's past held many species of animal now extinct**. This was shocking for some, since many people in Europe still believed all animals were created at the time of the Creation.

Fossil hunters found more fossils in the Jurakalk rock formation in the French Jura Mountains. Prussian geographer **Alexander von Humboldt** (1769–1859) identified the age of these limestone formations, thus setting a **basis for geologists to identify the Jurassic age**—the age of dinosaurs.

FRENCH MATHEMATICIAN PIERRE-SIMON LAPLACE (1749–1827) published his *Exposition du Système du Monde* (*The System of the World*), which included his theories on orbits and tides, and the **first fully scientific exposition of the nebular hypothesis**, first suggested by Immanuel Kant in 1755, which proposes that the Solar System began as a giant gas cloud (nebula) spinning around the Sun. It cooled and contracted to form planets.

For 19-year-old German mathematician, Carl Gauss, 1796 was a year of breakthroughs. On March 30, he proved that it was **possible to construct certain regular polygons**, including the 17-sided heptadecagon, with just a compass and a ruler—a problem that had eluded mathematicians since the time of Pythagoras. On April 8, he made key contributions to solving quadratic equations. On May 31, he created the **prime number theorem** to show how prime numbers are spread. And, July 10, he discovered that **every positive integer is representable as a sum of, at most, three triangular numbers**.

Since the 1720s, many had been protected against the deadly disease smallpox by inoculation, which involved deliberate infection with smallpox germs. But the inoculation had risks and could kill. In 1796, British country doctor Edward Jenner (1748–1823) noticed that dairymaids often failed to contract smallpox and began to wonder if they gained their immunity through exposure to a similar, milder disease known as cowpox. He thought it possible that **"vaccination" with cowpox would be safer than inoculation** with smallpox, and deliberately infected his gardener's son with cowpox. Vaccination originally meant inoculation with cowpox, but the term is now used for any inoculation by weak or killed germs. A few weeks later he attempted to infect the boy with smallpox, who proved to be immune and remained so all his life. Vaccination using a harmless form of a dangerous disease is now one of the most effective methods of disease prevention.



Edward Jenner vaccinating his son
Many people remained unconvinced about vaccination, even after it had been successfully performed in 1796. In 1797, Jenner vaccinated his 11-month-old son to prove that it was safe.

1795 French chef Nicholas Appert invents the **food-preserving jar**

April 7, 1795 French **gram is standardized**

1795 Prussian geographer Alexander von Humboldt studies the **Jurakalk rock** formation in the Jura Mountains

1795 French naturalist Georges Cuvier **identifies fossil bones** found in the Netherlands as belonging to a huge extinct reptile

November 30, 1795 British engineer Joseph Bramah patents **hydraulic machines**

December 13, 1795 Meteorite falls on Wold Newton, Yorkshire, England

Pierre-Simon Laplace develops a theory of **Solar System origin**

March 30 Carl **Gauss proves the heptadecagon** can be constructed with a ruler and compass

May 14 British doctor Edward Jenner demonstrates **vaccination**

May 31 German mathematician Carl Gauss proposes his **prime number theorem**

July 10 Gauss shows that every **positive integer** can be sum of at most three triangular numbers

First **cast iron aqueducts** completed in Britain